Unit 2

Lesson 2

Innovators in Action.

Code Red!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Understand the role of a vibration motor in an alert system and how it can be used to assist individuals with both hearing and vision impairments.
- Design an inclusive alert system by integrating a vibration motor, LED, and buzzer, ensuring it can alert people through sight, sound, and touch.

Challenge questions of the lesson.

- How do conditional statements (like if and else) transform an Arduino from a simple controller into a "smart" system capable of making decisions based on data?
- Imagine a smart home system. How would you use different conditional statements to ensure the lights turn on only when it's dark and someone is in the room, or if a specific time has passed?
- What are the key challenges in setting the exact "rules" or thresholds for your Arduino's decisions, and why is thorough testing of these rules so important for a reliable system?

SEL Relation

- **Self-Awareness:** Recognizing their own logical thinking skills and how they apply to programming.
- **Self-Management:** Practicing precision and patience when structuring complex code logic.
- **Social Awareness:** Understanding how automated decisions can affect users and communities, especially in safety systems.
- **Relationship Skills:** Collaborating effectively to debug complex code and explain logical processes.
- **Responsible Decision-Making:** Considering the ethical implications of automated choices and ensuring code serves the user's best interest.

Engage

Teacher's Role:

- **Set the Stage:** Guide students to think critically about the challenges faced by individuals with hearing or vision impairments. Ask open-ended questions to spark curiosity and reflection.
- **Facilitate Discussions:** Ask students to consider different forms of communication (visual, tactile, auditory) that could be used to alert individuals to danger. Encourage brainstorming and empathy in their responses.
- **Encourage Creativity:** Provide prompts for students to think of innovative ways to alert someone without relying on sound or sight.

Expected from Students:

- **Active Participation:** Engage in discussion about how to communicate a threat to someone who can't hear or see.
- **Brainstorm Ideas:** Students will suggest possible methods of communication (e.g., tactile alerts like vibration or visual signals like flashing lights).
- **Reflect on Real-World Needs:** Understand the challenges of communicating alerts to people with sensory impairments and propose viable solutions.

Parts of the book included.


Pyramakerz®

Let's Think!

If someone can't hear or see, how can we alert them to potential danger? What methods can we use to communicate the threat effectively?

Let's Think!

Objective:

Students will identify methods of alerting individuals who can't hear or see, considering both traditional and innovative methods for communicating danger effectively.

Possible Student Answers:

- Tactile signals (e.g., vibration motors, pressure sensors).

- Visual signals (e.g., flashing lights, colored indicators).
- Wearable devices (e.g., wristbands that vibrate when a threat is near).
- Use of universal signals (e.g., braille, large visual signs).

Activity Instructions:

1. Discussion Prompt:

Ask students, “If someone can’t hear or see, how can we alert them to potential danger?” Encourage students to think about real-world examples.

2. Brainstorm Solutions:

Students will suggest ways to alert individuals (e.g., vibration, flashing lights, wearable devices).

3. Sketch or Design:

Ask students to sketch or describe an alert system that could be used to alert someone who can’t hear or see.

4. Group Sharing:

Have students share their ideas in small groups and discuss how each solution would work in practice.

Possible Strategies for implementation:

- **Think-Pair-Share:** Have students first think individually about the problem, then pair up to discuss their ideas, and finally share with the class.
- **Questioning:** Ask follow-up questions like, “How would your idea work in a noisy environment?” or “What makes your solution effective for someone who is hard of hearing?”
- **Role-Play:** Have students role-play different scenarios where they must communicate a warning to someone without using sound or sight.

Explorer’s Toolkit

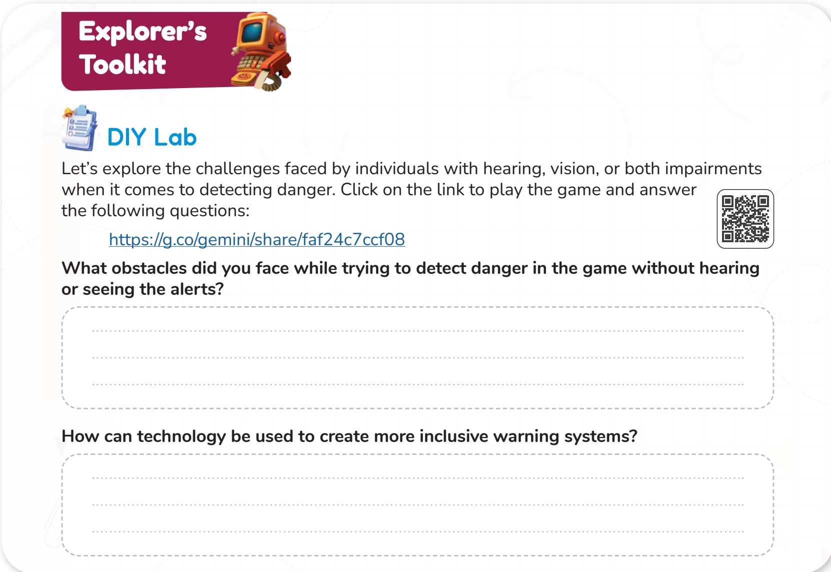
Teacher’s Role:

- **Set the Context:** Introduce the activity by explaining the challenges faced by individuals with hearing, vision, or both impairments in detecting danger.
- **Facilitate Engagement:** Guide students through the game, ensuring they understand the concept of non-verbal and non-visual alerts. Offer prompts during the game to keep students engaged and thinking critically.
- **Encourage Reflection:** After students finish the game, lead a discussion about the obstacles they faced and ask them to think about how technology could help make warning systems more inclusive.

Expected From Students:

- **Active Participation:** Students will play the game, paying attention to how the absence of auditory or visual cues affects their ability to detect danger.
- **Critical Thinking:** Students will reflect on the obstacles they encountered and propose solutions using technology to create more inclusive warning systems.
- **Discussion Contributions:** After completing the game, students will answer the reflection questions and engage in a class discussion about technology's role in accessibility.

Parts of the book included.



Explorer's Toolkit

DIY Lab

Let's explore the challenges faced by individuals with hearing, vision, or both impairments when it comes to detecting danger. Click on the link to play the game and answer the following questions:

https://g.co/gemini/share/faf24c7ccf08

What obstacles did you face while trying to detect danger in the game without hearing or seeing the alerts?

How can technology be used to create more inclusive warning systems?

DIY Lab

Objective:

Students will understand the challenges faced by individuals with hearing and vision impairments in detecting danger. They will reflect on their experience in the game and explore how technology can create more inclusive warning systems. Students will also consider how such solutions can improve safety and accessibility.

Activity link: https://g.co/gemini/share/faf24c7ccf08

Possible Answers:

- **Obstacles Encountered:**
  - Difficulty in detecting alerts (e.g., no sound or visual cues to indicate danger).
  - Inability to react quickly due to lack of sensory information.
  - Frustration from missing critical warnings.
- **Technology Solutions:**
  - Vibrating alarms or haptic feedback to alert individuals.
  - Use of flashing lights or visual indicators in strategic locations.
  - Wearable devices that send alerts through touch, vibration, or notifications.

Materials Needed:

- Computer or device with internet access to play the game.
- Notebook or device to take notes and record answers.
- Setup:
  - o Ensure each student has access to the game and can play it individually or in pairs.
  - o Provide time for students to reflect and take notes during and after the game.

Activity Instructions:

1. Play the Game.

Focus on detecting danger without relying on hearing or sight.

2. Take Notes:

As you play, jot down any obstacles you face in trying to detect danger without hearing or seeing the alerts.

3. Reflect:

After completing the game, answer the following reflection questions:

- o What obstacles did you face while trying to detect danger in the game without hearing or seeing the alerts?
- o How can technology be used to create more inclusive warning systems?

Possible Strategies for Implementation:

1. Pre-Watching: Set the Purpose

Ask students: "How do you think someone who can't hear or see would detect danger? What methods might be used to help them?" Encourage them to think about real-world applications of assistive technologies.

2. During the Game: Active Engagement

- **Strategy: "What Can You Feel?"**

  Have students focus on the challenges they face during the game, particularly how not being able to hear or see alerts affects their ability to react. Prompt students to note which sensory information they're missing and what could help in real life.

3. Post-Game Reflection: Group Discussion

- **Strategy: "Empathy Sharing"**

  After the game, encourage students to share their thoughts on how it felt to play without sight or sound. Ask: "How can we use technology to make alert systems more accessible for people with impairments?" Encourage students to brainstorm ideas like haptic feedback or visual alerts.

4. Collaborative Brainstorming

Allow students to work in pairs or small groups to come up with ideas for inclusive warning systems, considering the technology and tools available to them.

5. Wrap-Up: Connect to Real Life

- **Strategy: "Real-World Impact"**

  Finish the activity by asking students how they would feel if they relied on the technology they discussed. Prompt them to think about the impact of inclusive designs on real users and how these systems could save lives.

Explain

Teacher's Role:

- **Set the Context:** Introduce the video by explaining how vibration can serve as an alternative to sound or sight for alerting individuals to their surroundings, especially for those with hearing or vision impairments.
- **Facilitate Engagement:** Play the video and pause at key points to ask guiding questions, such as "How does vibration work as an alert system?" or "Why might vibration be more effective than sound for some people?"
- **Encourage Reflection:** After watching, prompt students to discuss how vibration could be incorporated into their own designs (e.g., smart alert systems).
- **Provide Support:** Offer assistance if students have trouble understanding how the vibration system is integrated into devices.

Expected From Students:

- **Active Watching:** Students should watch the video attentively, taking notes on how vibration is used to alert individuals to their surroundings.
- **Engage in Discussion:** After watching, students will reflect on the effectiveness of vibration alerts and consider how they can be used in their own projects (e.g., Smart Walker).
- **Apply Knowledge:** Students will suggest how vibration alerts could be incorporated into assistive technologies to improve accessibility and safety.

Parts of the book included.

Screen Lessons

Click the link to discover how vibration can be used to alert individuals to their surroundings:

https://www.youtube.com/watch?v=ZKoicU-zorA

Screen Lessons

Objective:

students will be able to:

- Understand how vibration can be used as an alert system for individuals with hearing or vision impairments.
- Analyze the potential applications of vibration in real-world assistive technology.
- Discuss the benefits and challenges of using vibration as a method of communication.
- Incorporate vibration alerts into their own Smart Walker design.

Resource Links:

Video Link: https://www.youtube.com/watch?v=ZKoicU-zorA

Possible Strategies for Implementation:

1. **Pre-Watching:**

Introduce the concept of vibration as an alert system. Ask students: "How do you think vibration could be used to help someone who can't hear or see?" Write answers on the board to engage their thinking.

2. **During Watching:**

Play the video, pausing at key points to prompt critical thinking:

- "What are some devices that use vibration alerts?"
- "How does the video show vibration as an effective alert?"

3. **Post-Watching:**

Discuss the video with students and ask them to reflect on the role of vibration in accessibility. Use prompts like:

- "How could vibration help a person with a hearing impairment?"
- "Where might we use vibration in the Smart Walker system?"

4. **Apply Learning:**

Encourage students to brainstorm how vibration can be incorporated into their own Smart Walker designs, focusing on its role in safety and accessibility

Elaborate

Teacher's Role:

- **Guide Understanding:** Review answers with the class, linking real-world examples to concepts like tactile feedback, energy transfer, and if conditionals.
- **Facilitate Exploration:** Introduce the Mastermind challenge and guide students to think about how to detect patterns and change alert behavior based on frequency.

- **Provide Support:** Help students with programming logic, using millis() and counters to track low-level signals and adapt alerts.
- **Encourage Iteration:** Remind students to test, refine, and improve their systems throughout the design process.
- **Foster Creativity:** Encourage creative thinking, experimentation, and troubleshooting. Reinforce that mistakes are part of learning.

Expected from students:

- **Active Participation:** Answer questions thoughtfully in the Apprentice activity and demonstrate understanding of core concepts.
- **Critical Thinking:** Design a system that detects patterns and adjusts alert behavior based on signal frequency.
- **Collaboration:** Work together, brainstorming ideas and solving problems with peers during the Mastermind activity.
- **Iteration:** Test systems, make adjustments, and refine logic based on results.
- **Reflection:** Reflect on challenges, explain design choices, and suggest improvements after completing activities.

Parts of the book included.

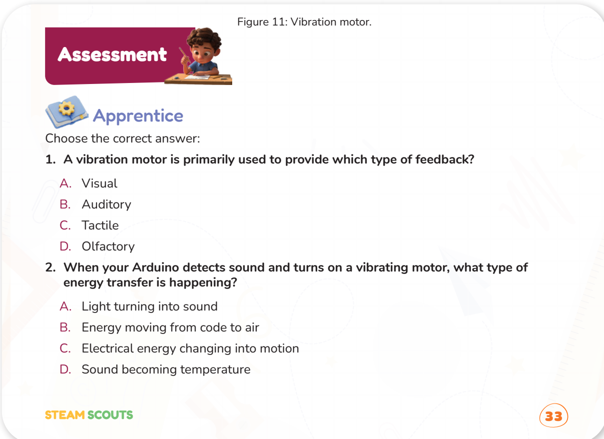


Figure 11: Vibration motor.

Assessment

Apprentice

Choose the correct answer:

1. A vibration motor is primarily used to provide which type of feedback?
   A. Visual
   B. Auditory
   C. Tactile
   D. Olfactory
2. When your Arduino detects sound and turns on a vibrating motor, what type of energy transfer is happening?
   A. Light turning into sound
   B. Energy moving from code to air
   C. Electrical energy changing into motion
   D. Sound becoming temperature

STEAM SCOUTS 33

Apprentice

Objective:

Students will demonstrate understanding of tactile feedback, energy transfer in motors, and correct usage of if conditional statements in Arduino code. They will apply knowledge from previous lessons and coding resources to answer conceptual and practical questions.

Answers:

1. **A vibration motor is primarily used to provide which type of feedback?**

Correct Answer: c. Tactile

2. **When your Arduino detects sound and turns on a vibrating motor, what type of energy transfer is happening?**

Correct Answer: C. Electrical energy changing into motion

3. **In your Arduino code, which of the following is the best way to tell the LED to turn red when the sound level is above 80?**

Correct Answer: A. if (soundLevel > 80) { turnRed(); }

Possible Strategies for Implementation:

- **Demonstration First**
  - o Show students a small vibration motor in action so they can feel and see the tactile feedback.
  - o Briefly explain each type of feedback (visual, auditory, tactile, olfactory) with real-life examples before they attempt Question 1.
- **Energy Transfer Example**
  - o Use a simple Arduino setup with a microphone sensor and a vibration motor.
  - o Trigger the motor when sound is detected so students can observe electrical energy turning into motion (for Question 2).
- **Code Walkthrough**
  - o Write a simple if statement on the board and explain each part:

    if (sensorValue > 80) → condition

    { activateVibration(); } → action when true
- Compare with incorrect options so they can understand why A is correct for Question 3.
- **Think-Pair-Share**
  - o Let students answer individually, then discuss with a partner before sharing with the class.
  - o Encourages peer teaching and deeper reasoning.
- **Hands-on Coding Practice**
  - o Provide a skeleton Arduino code where students fill in the condition to activate the motor at sensorValue > 80.
  - o Upload to an Arduino and test.
- **Link to Real-World Applications**
  - o Discuss how vibration motors are used in mobile phones for alerts, in wearables for fitness tracking, and in accessibility devices for people with hearing impairments.

Builder

Objective:

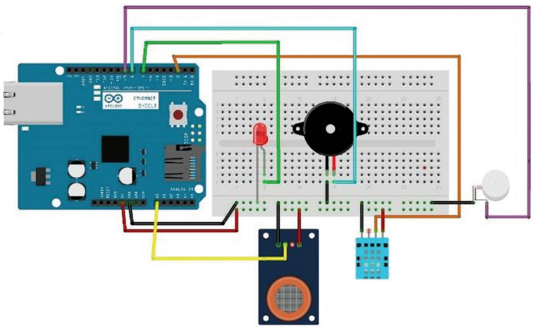
Students will integrate a vibration motor into their Fire Alert System, modify their Arduino code to trigger the motor based on sensor inputs, and refine their system for accurate alerting. They will apply previous knowledge of circuit design, coding, and sensors to build a functioning system that responds to gas and temperature readings.

Activity Instructions:

- **Introduce Vibration Motor:**

  Explain how vibration motors provide tactile feedback, offering an additional way to alert users to potential dangers, such as high gas levels or temperature.

- **Connect the Vibration Motor:**
  - o Connect the positive lead of the vibration motor to Arduino digital pin 3.
  - o Connect the negative lead to Arduino GND.
  - o (Optional: Use an NPN transistor and diode for more power if required for the motor.)



- **Modify the Code to Activate the Motor:**
  - o In the Arduino IDE, add the necessary code to turn on the vibration motor when gas levels or temperature readings exceed the set threshold.
  - o Example code:

cpp

Copy

```
if (gasLevel > gasThreshold || temp > 30) {
  digitalWrite(VIBRATION, HIGH);  // Turn on vibration motor
} else {
  digitalWrite(VIBRATION, LOW);   // Turn off vibration motor
}
```

- **Test the System:**
  - o After uploading the code, test the system by simulating high gas or temperature readings.
  - o Confirm that the vibration motor activates when the conditions are met and deactivates when the values return to normal.
- **Troubleshoot:**
  - o If the vibration motor doesn't activate, check the wiring and ensure the correct pin is referenced in the code.
  - o Verify that the sensor thresholds are set correctly for triggering the motor.
- **Refine the System:**

- o Once the motor is working, consider adjusting the threshold values to fine-tune the system for realistic alert conditions (e.g., gas level threshold of 400 or temperature of 30°C).

- **Test and Iterate:**
  - o Encourage students to test their systems in different scenarios and adjust the code or circuit if needed to improve performance or reliability.

Possible Strategies for Implementation:

- **Introduction and Conceptualization:**
  - o Discuss how vibration motors provide tactile feedback for people with hearing impairments. Use real-world examples (smartphones, controllers).
  - o Introduce tactile feedback as a key feature in assistive technology.
- **Modeling the Process:**
  - o Show how to connect the vibration motor to the Arduino and demonstrate the code to activate the motor based on sensor readings.
  - o Explain how conditional statements (if statements) trigger the motor when thresholds are exceeded.
- **Hands-On Exploration:**
  - o Allow students to connect their own motors and modify the code. Let them experiment with different threshold values.
  - o Use visual guides and step-by-step instructions to support students.
- **Collaborative Problem Solving:**
  - o Have students work in pairs to test and troubleshoot circuits. Encourage peer learning.
  - o Assign roles within the group (coder, tester, etc.) to build teamwork and collaboration.
- **Test and Iterate:**
  - o Guide students to test their circuits, adjust thresholds, and iterate the design. Remind them that engineering requires adjustments.
  - o Offer prompts like, “What happens when you change the threshold?”
- **Reflection and Review:**
  - o After testing, ask students to reflect on what worked, the challenges faced, and improvements they would make.
  - o Use questions like, "What changes would improve the design?"
- **Real-World Application:**
  - o Discuss how this technology could be used in real-world scenarios like smart homes or wearable devices for accessibility.
  - o Highlight how vibration feedback can be applied to improve devices for people with disabilities.

Arduino Code :

Here is the Arduino code that includes the vibration motor for your Fire Alert System with gas and temperature sensing:

cpp

Copy

```
#include <DHT.h>
#include <LiquidCrystal_I2C.h>
#include <Wire.h> // Make sure Wire library is included for I2C
#define MQ2_A0 A0        // Gas sensor analog output
#define DHTPIN 2         // DHT11 data pin
#define DHTTYPE DHT11
#define VIBRATION 3      // Digital pin for vibration motor
#define LED 4            // Visual alert LED
DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2); // Use the I2C address found by the scanner
int gasThreshold = 400;  // Tune this based on calibration
void setup() {
  Serial.begin(9600);
  Serial.println("Fire Alert System Initializing...");
  pinMode(MQ2_A0, INPUT);
  pinMode(VIBRATION, OUTPUT);
  pinMode(LED, OUTPUT);
  dht.begin();
  lcd.init();       // Initialize the LCD
  lcd.backlight();  // Turn on the backlight
  lcd.print("Initializing..."); // Initial message on LCD
  delay(1000);
  lcd.clear();      // Clear the initial message
  Serial.println("Initialization complete.");
}
void loop() {
  int gasLevel = analogRead(MQ2_A0);
  float temp = dht.readTemperature();
  // *** LCD Display ***
  lcd.setCursor(0, 0);
```

```
lcd.print("Gas: ");
lcd.print(gasLevel);
lcd.print("    "); // Add spaces to overwrite previous digits
lcd.setCursor(0, 1);
lcd.print("Temp: ");
lcd.print(temp);
lcd.print(" C   "); // Add spaces to overwrite previous digits
// *** Serial Monitor Output ***
Serial.print("Gas: ");
Serial.print(gasLevel);
Serial.print("  |  Temp: ");
Serial.print(temp);
Serial.println(" C");
// *** Alert Logic and LED Control ***
if (gasLevel > gasThreshold || temp > 30) {
  digitalWrite(VIBRATION, HIGH); // Activate vibration motor
  digitalWrite(LED, HIGH);       // Turn on alert LED
  Serial.println("ALERT! Gas level or temperature high!");
  lcd.setCursor(10, 0); // Move cursor to column 10 of the first line
  lcd.print("ALERT!");
} else {
  digitalWrite(VIBRATION, LOW);  // Deactivate vibration motor
  digitalWrite(LED, LOW);        // Turn off alert LED
  lcd.setCursor(10, 0); // Move cursor to column 10 of the first line
  lcd.print("OK    ");   // Display "OK" and clear any previous "ALERT!"
}
delay(1000); // 1 second delay
}
```

Explanation:

4. **Vibration Motor Control:**
   - The motor is connected to pin 3 (VIBRATION), and it is activated with digitalWrite(VIBRATION, HIGH) when either the gas level or the temperature exceeds the set thresholds. It is deactivated with digitalWrite(VIBRATION, LOW) when the values are safe.
5. **Alert Logic:**
   - The condition checks if the gas level (gasLevel > gasThreshold) or the temperature

(temp > 30°C) is above the defined threshold. If true, it activates both the vibration motor and the LED.

6. **LCD Display:**
   - Displays the current gas and temperature readings on the 16x2 LCD screen.
   - If the alert condition is triggered, it will display "ALERT!" on the LCD.
7. **Serial Monitor Output:**
   - Sends the gas level and temperature readings to the Serial Monitor for debugging and real-time monitoring.

Customizations:

- Threshold Adjustments:

You can modify the gasThreshold value to calibrate the gas sensor for your specific environment. Similarly, adjust the temperature threshold (currently set at 30°C).

- Pin Assignments:

Ensure the vibration motor is connected to pin 3, the LED to pin 4, and the gas sensor and temperature sensor pins are correctly connected according to the defined pin numbers.

Mastermind

Objective:

Students will design and program a smart alert system capable of recognizing repeating low-level sensor signals over time. They will develop a system that adapts its alert behavior based on the frequency of these signals, enhancing the system's ability to detect growing problems through pattern recognition.

Answer:

Approach:

Instead of only checking if a sensor value crosses a high threshold once, the system will count how many times the value crosses a lower threshold within a given period. If this count is high enough, it will trigger an alert.

Example Scenario:

- Gas sensor reading above 50 is considered a "low-level warning."
- If this happens 5 or more times in 10 minutes, activate the vibration motor or buzzer.

Arduino Example Code:

cpp

CopyEdit

```
int sensorPin = A0;      // Sensor input pin
int sensorValue = 0;
int alertCount = 0;
unsigned long startTime = 0;
```

```
unsigned long timeWindow = 600000; // 10 minutes in milliseconds
void setup() {
 Serial.begin(9600);
 startTime = millis();
}
void loop() {
 sensorValue = analogRead(sensorPin);
 // If low-level signal detected
 if (sensorValue > 50) {
  alertCount++;
  delay(200); // Small delay to avoid counting the same event repeatedly
 }
 // Check if time window passed
 if (millis() - startTime >= timeWindow) {
  if (alertCount >= 5) {
   Serial.println("Alert! Repeated low-level signals detected.");
   // activateVibration(); // Example action
  }
  // Reset counter and timer
  alertCount = 0;
  startTime = millis();
 }
}
```

Activity Instructions:

1. **Introduce the Concept**
   - Explain that many alert systems only react when a threshold is crossed.
   - Use simple real-world examples (e.g., car maintenance reminders, daily step goals in fitness trackers, or phone storage warnings).
   - Discuss why repeated small alerts might still be important (e.g., detecting a small gas leak before it becomes dangerous).
2. **Clarify Key Vocabulary**
   - Threshold: A set limit that triggers an action.
   - Low-level signals: Small or infrequent readings that might not seem dangerous at first.
   - Patterns over time: Trends or repeated occurrences.
3. **Demonstrate the Problem**
   - Show a basic Arduino (or pseudo) example that only triggers once the value is above a set number.
   - Ask: "What if the value is low but happens often? Should the system still respond?"
4. **Explain the Challenge**
   - Students will work at home to research and design a way for a system to recognize repeated signals.
   - Emphasize they should think about counting occurrences and tracking over time (e.g., using a counter variable or a timer).
5. **Provide Research Guidance**
   - Suggest search terms: Arduino count occurrences, detect repeated events over time, pattern detection with sensors.
   - Encourage them to find examples or tutorials online and adapt them.
6. **Set Expectations for Submission**
   - Students should return with:
     - A brief explanation of their approach (can be written or drawn).
     - Optional: Sample code or pseudocode showing how the repetition logic works.
7. **Encourage Creativity**
   - Remind students they can choose any type of sensor (sound, temperature, gas, motion, etc.) for their example.
   - They can simulate the concept if they don't have actual hardware at home.

Evaluate

Teacher's Role:

- **Objective:** Encourage students to take their learning from the Explain and Elaborate phases and apply it in designing a practical, inclusive solution. The goal is to design a flexible band that can be worn by individuals with disabilities to receive emergency alerts.
- **Facilitate creativity:** Allow students to imagine how such a band could work in real-life scenarios. Encourage them to consider accessibility features like vibration, sound, and color indicators for various alert types.
- **Provide resources:** Show students the Thingiverse link to the 3D model that can be printed, but encourage them to brainstorm and customize their designs further.
- **Foster collaboration:** Allow students to work in pairs or small groups, sharing ideas and providing feedback to enhance the designs.
- **Assessment guidance:** Help students focus on the practicality, accessibility, and creativity of their design solutions.

Expected from Students:

- **Imagination and Design:** Students are expected to creatively design a flexible band that could be used by people with disabilities to receive emergency alerts. They should consider factors like comfort, material, ease of use, and functionality.
- **Use of Resources:** Students should refer to the 3D model on Thingiverse for inspiration, and can use it to base their designs, but they are also encouraged to make improvements or customize it according to their ideas.
- **Documentation and Reflection:** Students will need to document their design process, explaining how their band works, how it caters to different disabilities, and any unique features they've added.
- **Presentation:** Students will present their design, explaining how it works, who it's for, and how it could benefit individuals with disabilities in emergency situations.

Showcase

Objective:

Design a flexible band that people with disabilities can wear to receive alerts during emergencies. This project allows students to apply the concepts they've learned about accessibility, technology, and emergency preparedness in a real-world context.

Activity instructions:

1. **Step 1: Brainstorm and Sketch**

   - Begin by discussing the key needs of individuals with disabilities during an emergency. Consider the types of alerts (e.g., vibration, flashing lights, sound) that may be necessary for those with different impairments (e.g., hearing, vision).

- Sketch the design of your flexible band. Think about what materials it could be made from (e.g., rubber, soft plastics) and how it could deliver alerts (e.g., vibrations, LEDs, sound).
- Write down any additional features or customizations that make your band more inclusive.

2. **Step 2: Design and Model Your Band**

- Using a 3D modeling tool (such as Tinkercad), begin designing your flexible band. You may use the model from Thingiverse as a reference, but your design should incorporate your own ideas and creativity.
- Focus on ensuring that the band is adjustable, comfortable, and easily worn by individuals with different disabilities.

Thingiverse Reference

You can download the initial model from: https://www.thingiverse.com/thing:6969175

This can be used as a starting point for your design or inspiration.

3. **Step 3 :3D Printing the Model**

- Once your design is complete, save it and prepare the file for 3D printing. Make sure the dimensions and flexibility of the band are suitable for a real-world application.
- Print the model using a 3D printer, or submit the design to your teacher or a print lab.

4. **Step 4: Reflection and Evaluation**

- After printing, reflect on how well your design meets the needs of people with disabilities.
- Consider how your band could be improved. What other features could be added to make it more useful? Would it work in different environments? Is it easy to use?
- Prepare a short presentation where you explain your design, its purpose, and the key features. Be sure to address why your design is inclusive and accessible.

Possible Strategies for Implementation:

- **Design Flexibility:** Encourage students to think about how the design could be adapted for different needs. For instance, how could the band be used for someone who is hearing-impaired? Or for someone who is visually impaired?
- **Peer Review:** Have students share their designs with a partner for feedback. Discuss any improvements that could be made and how their design could better meet the needs of users.
- **Showcase Presentations:** Allow students to present their designs in front of the class. This gives them an opportunity to showcase their creative process and receive constructive feedback.

Rubric with Clear Criteria

Criteria	Needs Improvement (1–2)	Satisfactory (3)	Excellent (4–5)
Design Creativity	Minimal creativity, lacks original features.	Creative, but lacks some essential accessibility features.	Highly creative with unique features and great attention to accessibility.
Functionality	Band design does not meet basic functionality or accessibility needs.	Band is functional, but may not be fully inclusive or practical.	Fully functional, with features designed for different disabilities.
Presentation and Reflection	Limited explanation of design features.	Clear explanation with some details on why it works.	Clear, detailed explanation with thoughtful reflection on design improvements.
Practicality	Design lacks real-world applicability.	Design could work with minor adjustments.	Design is highly practical and easily usable in real-life situations.

Don’t forget to go back to EDP paper and add this to improve part:

Improve

We designed and built a wearable alert band by connecting sensors, an LED, a buzzer, and a vibration motor to an Arduino. The vibration motor was embedded into a custom 3D-printed wristband so it touches the skin and provides tactile feedback. Our Arduino code monitors the sensor readings, and when they cross a set safety threshold or occur repeatedly over time, the band vibrates to alert the wearer. This allows the user to receive immediate warnings without needing to look at a screen or hear a sound, making it especially useful in noisy environments or for people with hearing impairments.

Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. **Introduce the Reflection**
   - Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."
2. **Guide Students Through the Checklist**
   - Read each statement aloud one at a time.
   - After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
   - Example prompts:
     - "Did you learn how people share ideas without talking?"
     - "Do you know how tools can help people talk?"
3. **Student Completion**
   - Let students check off or color the "Now I Can" boxes they feel good about.
   - Remind them: "It's okay if you didn't check all of them. We're all still learning."
4. **Optional Extension**
   - Ask students to circle the one they're most proud of.
   - Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. Next Generation Science Standards (NGSS):
   - **3-5-ETS1-1:** Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.
   - **4-PS3-4:** Apply scientific ideas to design, test, and refine a device that converts energy from one form to another (e.g., electrical energy activating light, sound, or vibration).
   - **3-PS2-1:** Plan and conduct investigations to provide evidence of the effects of balanced and unbalanced forces on the motion of an object — supporting understanding of how sensors detect and respond to changes.
2. International Society for Technology in Education (ISTE) Standards for Students:
   - **1. Empowered Learner:** Use technology to take an active role in choosing, achieving, and demonstrating competency in their learning goals.

- **4. Innovative Designer:** Use a variety of technologies within a design process to identify and solve problems by creating new, useful, or imaginative solutions.
- **6. Creative Communicator:** Communicate clearly and express themselves creatively for a variety of purposes using digital tools.

3. Common Core State Standards (CCSS) — English Language Arts:

- **CCSS.ELA-LITERACY.SL.3.1:** Engage effectively in collaborative discussions on grade-appropriate topics and texts, building on others' ideas and expressing their own clearly.
- **CCSS.ELA-LITERACY.W.3.2:** Write informative/explanatory texts to examine a topic and convey ideas clearly.
- **CCSS.MATH.PRACTICE.MP4:** Model with mathematics — interpret data and apply conditions logically in designing alert systems.

4. Social-Emotional Learning (SEL):

- **Self-Awareness & Responsible Decision-Making:** Understanding the importance of creating inclusive and reliable alert systems that protect and assist all users, including those with sensory impairments.